

3. Rebalance!

Mr. Market is special in another way: *the average investor never rebalances*. As Chapter 4 explains, for everyone who rebalances by buying low and selling high, there must be someone on the other side who loses money by buying high and selling low. During the financial crisis, CalPERS did a terrible job at rebalancing and sold equities when equity prices were at their lowest, and expected returns were at their highest.²⁵ Who was on the other side? Norway was the world's biggest buyer of equities in the last quarter of 2008.

Rebalancing is tricky because, for dynamic factors, you don't necessarily want to rebalance to dollar positions. In fact, for strict long-short dynamic factor positions, your dollar position is zero! But you want to rebalance as it forces you to cut your exposure when things have gone well, and add exposure in the opposite case, bringing you back to your optimal exposure to dynamic factors. You need to decide on rebalancing to risk exposures, volatility weights, exposures determined by contributions to risk or value-at-risk, etc. (I cover some of these cases in Chapter 4.) Pick a set of (risk) weights, then rebalance.

If you're good at rebalancing, you're ready for *factor timing*. Many factors are predictable (although the degree of predictability is very small, see Chapter 8), and you can exploit mean reversion in factor returns by incorporating valuation information in your investment process. You want to add more factor exposure beyond what rebalancing implies when those factors are cheap. Rebalancing already forces you to buy low and sell high; with factor timing you can buy even more when factor strategies become dirt cheap.

In my view of factor investing, *it is optimal for investors to hold non-market weighted portfolios*. Traditional asset management, as exemplified by Norway, (often unwittingly) takes on large amounts of factor risk. An influential article by Charles Ellis (1975) calls active management a "loser's game" because for every winner there is a loser and after transactions costs, there are only losers, on average. Active management can be a (relatively expensive) way to access factors, and it is optimal for investors to have different factor exposures. Active management, expressed through factor risks, may not be a loser's game. While for every person on one side of a factor trade, there is a person on the other side, both investors are happy because they improve their risk–return profiles by offsetting factor positions. For example, certain investors are happy to forego the premium for selling volatility to avoid periodic crashes. Instead, they purchase protection that pays off when volatility spikes. This makes everyone better off.

Investors find rebalancing standard stock-bond positions hard, and rebalancing dynamic factor positions is even harder. As Chapter 4 discusses, behavioral tendencies make investors most likely to give up on factor strategies after a string of

²⁵ See "California Dreamin': The Mess at CalPERS," Columbia CaseWorks ID#120306 and Chapter 15. Also see Ang and Kjaer (2011).